

Hi, I'm Jonathan.

I'm a **3rd-year Electrical Engineering** student at Carleton University.

I'm passionate about **hardware, wireless systems, robotics** and **artificial intelligence**

A bit about myself

I grew up in Ottawa with a longstanding interest in how things work at a physical and mechanical level. That curiosity shapes how I approach engineering throughout my studies and personal projects. I like taking an idea from a rough sketch to something that actually works, testing it until it breaks or reaches its limit, figuring out why, and then fixing it properly. I've picked up a broad set of practical skills that way and constantly try to gain more experience in whatever ways I can.

Outside of school, I work as a lifeguard, swim instructor, and first aid instructor with the City of Ottawa. I have coached and competed nationally in competitive speed skating, which has helped me build up a good set of interpersonal and self-development skills through both the work and sport avenues. I also have an interest in personal finance and economics, which has grown into a broader curiosity about practical AI and where it genuinely fits into engineering and finance, rather than the superficial hype around it.

Check out my builds!

A growing list of different projects and challenges that I have tackled - click "View Project →" on any card for the full story, photos, and what went wrong along the way.

Tools & technical proficiencies

Design

Altium Designer
KiCad
Autodesk Fusion CAD
Component Selection
Breadboarding

Programming

C / C++
Verilog HDL
Python
Jupyter Notebook
Java

Git Version Control

Platforms, Testing & Simulation

Arduino

Raspberry & Orange Pi (Linux)

STM & ESP 32

FPGAs & AMD Vivado

Oscilloscope / Multimeter Skills

What I'm keeping an eye on

Cutting-Edge Robotics

Legged and humanoid robots closing the gap between lab demos and real-world deployment. I'm most interested in advancements with actuator technology and integrations with machine learning.

RF & Space Communications

6G and IoT requirements are pushing wireless communication to its limits. I have a fascination for satellite constellations and am keen to understand antenna designs and endless uses for wireless communication.

AI & the Road to AGI

Looking past the buzzwords, I am excited to see the real use cases for AI systems within certain sectors like transport, industry, chip design and global finance. I am trying to embrace tools as they become available and learn how they can become powerful personal and corporate systems.

Let's connect.

I'm currently looking for a 2027 Summer Co-op placement. Always happy to talk about a project, a role, or just electronics and technology in general.